

Technical Report for NIE-ROZ Seminar Work

Melanoma Classification with Dominant Local Binary Patterns

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Abstract

This report describes a texture-based pipeline for melanoma recognition in dermoscopic images, inspired by *Dominant Local Binary Patterns for Texture Classification* (DLBP) and complemented with Normalized Gabor Features (NGF). Texture descriptors are computed inside a lesion mask and a non-linear SVM (RBF kernel) is used for classification. The system is evaluated on two train/test splits of a 7 000-image demo subset from the ISIC 2020 Challenge dataset, obtaining an accuracy (hit rate) close to 98%.

1 Introduction

In dermoscopy, visual clues linked to melanoma often appear as texture and micro-structure: pigment networks, globules, and irregular streaks tend to repeat locally, while broader patterns may extend over larger areas. A practical way to capture both aspects is to combine (i) local descriptors that are relatively robust to illumination changes and (ii) multi-scale frequency cues.

The pipeline discussed here follows this intuition. Dominant Local Binary Patterns are used to emphasize the most representative local codes, limiting the influence of rare and noisy patterns. Normalized Gabor Features are then added to provide complementary information at coarser scales, where LBP-style descriptors may be less informative.

2 Goal

Given a dermoscopic RGB image, the objective is to predict whether the lesion corresponds to malignant melanoma or to a benign lesion. Performance is measured primarily through classification accuracy (hit rate) on held-out images.

3 Dataset

Images are taken from the ISIC 2020 Challenge archive. The complete collection contains 33 126 images; for development and demonstration, a subset of 7 000 images (“demo dataset”) is considered. Each image is associated with a binary segmentation mask that separates lesion pixels from background skin (Fig. 1). The mask is used to restrict feature extraction to the lesion area.

3.1 Train and Test data

Multiple train/test ratios were used during development. In this report, two representative splits are discussed: 1:1 and 7:3. The first provides a balanced evaluation protocol, while the second prioritizes training data availability and offers an additional reference point for stability across splits.

4 Method

4.1 Preprocessing

Grayscale conversion. LBP descriptors are defined over intensity values. For this reason, the RGB image is converted to grayscale before computing local texture codes.

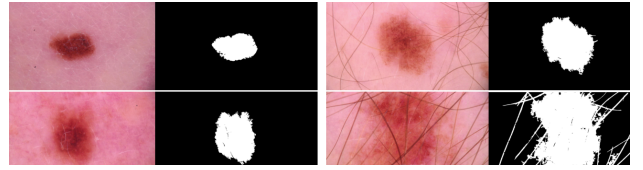


Figure 1: Examples of dermoscopic images (left in each pair) and corresponding binary lesion masks (right). The mask is used to limit feature extraction to lesion pixels.

Lesion masking. A segmentation model provides a binary lesion mask. Feature extraction is performed only on lesion pixels, reducing the influence of background skin texture, hair, and illumination artifacts (see Fig. 1).

4.2 Feature extraction

Rotation-invariant LBP. For each pixel, an LBP code is computed by comparing the center intensity with P neighbors sampled on a circle of radius R . Rotation invariance is obtained by circularly rotating the binary string and selecting the minimum value among rotations. A histogram of rotation-invariant codes is accumulated over the lesion area.

Dominant LBP (DLBP). Rather than using the full LBP histogram, DLBP keeps only a restricted set of patterns. Specifically, the feature vector is built from counts of the K most frequent rotation-invariant patterns, while less frequent codes are discarded. This reflects the empirical observation that real textures are often governed by a limited number of stable local configurations, and it helps suppress the contribution of rare/noisy patterns.

Normalized Gabor Features (NGF). To capture structures beyond the immediate neighborhood of LBP, Gabor filter responses at multiple scales and orientations are computed. Responses are aggregated (e.g., through average magnitude over the lesion) and normalized to mitigate global contrast differences. The final descriptor is obtained by concatenating DLBP and NGF components.

4.3 Classifier

Classification is performed using a Support Vector Machine with RBF kernel. The RBF kernel enables a non-linear

Split (train:test)	Acc.	Prec.	Rec.	F1
1:1 (3500/3500)	0.983429	0.98371	0.999710	0.991645
7:3 (4900/2100)	0.983810	0.98381	1.000000	0.991839

Table 1: Performance on the demo subset with SVM-RBF. Accuracy corresponds to the hit rate.

separation in feature space, which is typically beneficial when benign and malignant lesions share similar low-level appearance but differ in more subtle combinations of cues.

4.4 Implementation details and normalization

Choice of K for DLBP. A common strategy is to select K so that the retained patterns cover most of the histogram mass observed on the training set (for instance, around 80% cumulative frequency). Once fixed, the same set of dominant patterns is used for feature extraction on test images, ensuring consistent dimensionality and interpretation.

NGF normalization. Let m_i denote the average magnitude response of the lesion region to the i -th Gabor filter. Normalized features can be written as

$$g_i = \frac{m_i}{\sum_j m_j},$$

which reduces sensitivity to global intensity scaling and makes different images more comparable in terms of relative frequency content.

Feature scaling for SVM. Before training, each feature dimension is standardized (z-score) using training-set statistics only. This step prevents dimensions with larger numeric ranges from dominating RBF distances.

Computational aspects. For fixed P and R , the LBP/DLBP computation is linear in the number of pixels. Histogram operations dominate the runtime, while the selection of dominant codes is performed over a bounded histogram size, keeping the overall pipeline relatively lightweight.

5 Experiments

Performance is reported for two dataset splits on the 7 000-image demo subset, always using the same pipeline (masking + DLBP + NGF + SVM-RBF). Table 1 summarizes accuracy (hit rate) and additional metrics.

6 Discussion

The results show very high recall (approximately 1.0) on both splits, meaning melanoma cases are rarely missed under this protocol. Precision is slightly lower, suggesting that some benign lesions are flagged as melanoma, a behavior that is common in safety-oriented screening settings.

Masking is important: by restricting computation to lesion pixels, background-driven patterns (skin texture, hair, illumination gradients) have less influence on both DLBP histograms and Gabor responses (Fig. 1). In addition, DLBP provides a compact representation by focusing on dominant local configurations, while NGF contributes information about larger-scale organization that may not be fully captured by local neighborhoods.

Why masking helps. Without masking, stable background structures can produce consistent local codes that inflate the histogram and bias the classifier toward non-lesion cues. When the feature extraction is confined to the lesion, the descriptor becomes more closely tied to diagnostically relevant texture (networks, irregular edges, streak-like structures), and NGF tends to describe lesion-scale frequency content rather than background contrast changes.

Typical failure modes. False positives are expected when benign lesions present strong irregular texture (e.g., dense globules or high-contrast networks). False negatives can occur for low-contrast lesions, very small melanomas, or when the segmentation mask underestimates the lesion boundary and removes informative regions.

On combining DLBP and NGF. DLBP mainly captures fine micro-texture, whereas NGF emphasizes more global frequency structure. The concatenation is therefore expected to be more robust than either component alone, especially when lesions differ in both local detail and coarse organization.

Finally, the reported numbers should be interpreted in context: only a reduced demo subset is considered and the evaluation relies on two hold-out splits. A more complete assessment would include ablations (DLBP-only vs. NGF-only vs. combined) and a broader evaluation protocol on the full dataset.

7 Conclusion

A texture-driven melanoma classifier based on lesion masking, DLBP+NGF feature extraction, and an RBF-SVM classifier has been presented. On a 7 000-image demo subset of ISIC 2020, an accuracy of approximately 98.4% is obtained on two representative train/test splits.

References

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